
Cover time of options in one-dimensional environments

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Abstract

Effective exploration is crucial for the success of reinforcement learning. However, in practice, exploration is often inadequately addressed, with many applications relying on agents that explore the state space by taking uniformly random actions. We first observe that in a simple one-dimensional environment (a chain), the expected number of episodes for a random agent to reach the terminal state grows exponentially, specifically $\mathcal{O}(2^N)$, where N is the size of the chain. In contrast, by learning options that make use of knowledge acquired in previous episodes to explore more effectively in subsequent episodes. We demonstrate that, with options, an agent requires only $\mathcal{O}(\sqrt{N})$ episodes to reach the terminal state, offering a significant improvement in exploration efficiency.

Keywords: Random walks, Markov Decision Processes, Reinforcement Learning

Acknowledgements

1 Introduction

Effective exploration is a fundamental component of successful reinforcement learning. In practice, however, exploration is frequently underemphasized, with many applications relying on agents that explore states by taking uniformly random actions. We begin by noting that in a simple, one-dimensional environment—specifically, a chain—the expected number of episodes required for a random agent to reach the terminal state is $\mathcal{O}(2^N)$, where N denotes the size of the chain, as similarly noted by Osband et al. [2016].

Options Sutton et al. [1999] are temporally extended actions that can be reused and offer more efficient exploration compared to the commonly employed random exploration strategy. Specifically, we envision the instantiation of the agent through the use of *eigenoptions*, which are options derived from the eigenvectors of the Laplacian or successor representation. These eigenoptions effectively embody the theoretical advantages of exploration by reusing learned knowledge, thereby accelerating the learning process.

We demonstrate that by leveraging options that build on previous episodes to make progressively more informed decisions, an agent requires only $\mathcal{O}(\sqrt{N})$ episodes to reach the terminal state. Our work provides rigorous theoretical support for the utility of options in exploration, contrasting with previous research, which has primarily provided practical evidence. In doing so, we offer new insights into how structured exploration strategies can significantly enhance the efficiency of reinforcement learning agents.

2 Preliminaries

We consider a problem involving a chain of $N + 1$ states, where each state is enumerated in the set $[N] = \{0, \dots, N\}$. At each state, an agent has two available actions: -1 and $+1$. In state 0, both actions lead to state 1. For a state i where $1 \leq i \leq N$, action -1 leads to state $i - 1$ and action $+1$ leads to state $i + 1$.

The agent interacts with the environment in episodes, each consisting of N steps, starting from state 0. If the agent selects actions randomly at each time step, the probability of reaching the last state, N , in a single episode is $\frac{2}{2^N}$, which simplifies to $\frac{1}{2^{N-1}}$, and can be written as $2 \cdot \frac{1}{2^N}$. Consequently, the expected number of episodes required for the agent to reach state N is $\frac{1}{2} 2^N$, which is $\mathcal{O}(2^N)$. This quantity grows exponentially in N , indicating that the expected number of episodes to reach the terminal state becomes exceedingly and prohibitively large as N increases.

We summarize this result in the following remark:

Remark. The expected number of episodes for a random agent to reach state N in a chain of $N + 1$ states is $\frac{1}{2} 2^N$.

3 Result

Alternatively, suppose that the agent, endowed with the capacity to learn and reuse options Sutton et al. [1999], begins each episode by returning to the state where it finished the previous episode and avoids taking actions that would move it backwards from that state. More specifically, assume that after the first episode, in which the agent takes random actions, the agent finishes at state n . In the second episode, the agent first reaches state n by selecting the action $+1$ exactly n times, then continues with random actions, with the condition that once at state n , it selects action $+1$ again.

At the end of the first episode, the expected state of the agent is $\sqrt{\frac{2}{\pi}}\sqrt{N}$ Grünbaum [1960]. At the end of the second episode, the expected state is given by the sum:

$$\sqrt{\frac{2}{\pi}}\sqrt{N} + \sqrt{\frac{2}{\pi}}\sqrt{N - \sqrt{\frac{2}{\pi}}\sqrt{N}}.$$

In general, let Y_t denote the state where the agent is expected to finish episode t . The expected state at the end of episode $t + 1$ is given by the recursive relation:

$$Y_{t+1} = Y_t + f(Y_t),$$

where $f(y) = \sqrt{\frac{2}{\pi}}\sqrt{N - y}$.

This recursive relation is the first-order Euler method with step size $h = 1$ for solving the ordinary differential equation:

$$\dot{Y}(t) = f(Y(t)),$$

with the boundary condition $Y(0) = 0$. We can use this ordinary differential equation to approximate the behavior of the discrete Euler method governing the expected walk Y_t of the agent. The closed form solution to the ordinary differential equation is:

$$Y(t) = \frac{(2\sqrt{2\pi}\sqrt{N} - t)t}{2\pi},$$

Solving $Y(t) = N$ for t in an attempt to find the episode at which the agent is expected to reach the terminal state N , we obtain:

$$t = \sqrt{2\pi}\sqrt{N}.$$

We will show that $Y_t > Y(t)$ for all t , and consequently, the state where the agent is expected to be at the end of episode t is greater than $Y(t)$. Therefore, the agent is expected to reach state N less than $\sqrt{2\pi}\sqrt{N}$ episodes.

For the base case, we have $Y_1 = Y_0 + f(Y_0)$, where $Y_0 = 0$, and hence:

$$Y_1 = \sqrt{\frac{2}{\pi}}\sqrt{N}.$$

We also have the solution to the ordinary differential equation at $t = 1$:

$$Y(1) = \frac{(2\sqrt{2\pi}\sqrt{N} - 1)1}{2\pi} = \sqrt{\frac{2}{\pi}}\sqrt{N} - \frac{1}{2\pi}.$$

Since $Y_1 > Y(1)$, the base case holds. For the induction step, we assume that $Y_t > Y(t)$ holds for some t . Then, for episode $t + 1$, we have:

$$Y_{t+1} = Y_t + f(Y_t).$$

By the induction hypothesis, $Y_t > Y(t)$, and since the function $g(y) = y + f(y)$ is monotonically increasing, it follows that:

$$Y_{t+1} = g(Y_t) > g(Y(t)) = Y(t) + f(Y(t)).$$

Since the function f is concave, this quantity is greater than $Y(t + 1)$, completing the induction step.

We summarize the result in the following theorem:

Theorem 1. The expected number of episodes for the expanding agent to reach state N is smaller than $\sqrt{2\pi}\sqrt{N}$.

The implication of the result is that, while acting randomly the agent is expected to take $\mathcal{O}(2^N)$ episodes to reach the terminal state, endowed with the capacity to learn and reuse options, the agent is expected to take less than $\mathcal{O}(\sqrt{N})$ episodes, a significant reduction.

4 Conclusion

In this work, we demonstrated that in a chain of $N + 1$ states, a random agent is expected to take $\mathcal{O}(2^N)$ episodes to reach state N . In contrast, we showed that leveraging the concept of options, the agent is expected to reach state N in $\mathcal{O}(\sqrt{N})$ episodes, offering a significant improvement in exploration efficiency.

Future work could extend these results to environments with higher dimensions, exploring how the benefits of structured exploration, such as the use of options, scale in more complex settings.

References

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